



Miguel Ricardo
González Márquez
ricglez90@gmail.com
(633) 115-3923

Address
Calle Roma #336
Ensenada, Baja
California 22785
Mexico

Links
Github
LinkedIn
Google Scholar

Miguel Ricardo González Márquez

Ph.D. in Sciences | Lead Computer Vision Engineer

Summary Senior AI Developer and Computer Vision Architect with a Ph.D. in Sciences. Core expertise spans deep learning inference optimization using **C++**, **CUDA**, **PyTorch**, and **TensorRT** to build robust, production-ready vision pipelines. Proven ability to architect complex end-to-end systems, from mobile edge photo-capture and depth map segmentation to scalable cloud dashboards powered by **Node.js** and **Vue.js**. Brings applied experience integrating LLMs and agentic workflows (LangChain, RAG) to augment multimodal platforms. With extensive pedagogical experience as a university professor, I offer a unique capability to mentor engineering teams, establish rigorous technical standards, and lead cross-functional projects from conceptual design to deployment.

Education

2007-2011, B.S. in Computer Systems Engineering

Instituto Tecnológico de Agua Prieta (TecNM/ITAP)

2016-2018, M.S. in Engineering

Universidad Autónoma de Baja California (UABC)

2018-2023, Ph.D. in Sciences

Universidad Autónoma de Baja California (UABC)

Professional Experience

March 2022 - Present, Lead AI Researcher & Developer, Samsung Research Tijuana

Lead researcher developing advanced computer vision solutions for image and video processing. Promoted to Project Lead in 2022. Responsible for architecting inference pipelines, optimizing deep learning models for production environments, and guiding the engineering team through the full AI product lifecycle. Serve as a core technical mentor, actively training new team members to successfully pass rigorous internal programming skills certifications.

January 2015 - June 2016, IT Support & Network Technician, Instituto Tecnológico de Agua Prieta

Provided technical support for campus IT infrastructure; installed, maintained, and configured the campus network; developed and implemented institutional software projects.

Teaching & Mentorship Experience

August 2024 - Present, Adjunct Professor, CETYS Universidad Ensenada

Lecturer for Software Engineering and Digital Graphic Design Engineering programs. Designing and delivering project-based curricula focused on modern full-stack architectures, API integrations, and scalable front-end frameworks.



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Communication & Leadership

2025, *Panelist*, 4th Colloquium on Applications of Artificial Intelligence in Science

Roundtable panelist discussing the integration of AI in scientific research.

2025, *Speaker*, Samsung Week - INOVATEC - Instituto Tecnológico de Tijuana

Conference presentation: "Image Generation with Artificial Intelligence."

2024, *Speaker*, INOVATEC - Instituto Tecnológico de Tijuana

Conference presentation: "The Necessity of Lightweight AI Models."

2024, *Speaker*, DIGEEK 2024 CETYS Ensenada

Keynote presentation: "New AI Technologies."

2020, *Oral Presentation*, ROPEC 2020 (Ixtapa, Guerrero)

Virtual presentation at the 22nd IEEE Autumn Meeting on Power, Electronics and Computing.

2016, *Instructor*, COMEST 2016

Delivered a comprehensive Web Development course at Instituto Tecnológico de Agua Prieta for Computer Systems Engineering students.

Technical Skills

Programming Languages

- Python
- C++
- JavaScript
- R
- Matlab

Web & Cloud Architecture

- Node.js
- Vue 3 (Vite)
- Supabase
- Express.js
- API Dev

Machine Learning & Computer Vision

- PyTorch
- TensorRT
- ONNX Runtime
- CUDA
- LangChain
- OpenCV
- Scikit-Learn
- Matplotlib
- Agentic AI

Awards

2018, *Academic Merit Award*, UABC

Awarded for achieving the highest GPA in the Master's in Engineering program.

2011, *Valedictorian*, TecNM/ITAP

Awarded for achieving the highest GPA in the Computer Systems Engineering graduating class.

Interests

Professional

Passionate about bridging the gap between academia and the tech industry. Focused on applied research, advanced computer vision, deep learning inference optimization, and mentoring the next generation of software engineers to solve real-world industrial challenges.



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Scientific Publications

Comparison of CNN architectures for single grape detection

<https://doi.org/10.1016/j.compag.2025.109930>

Multiple circle detection in images: a simple evolutionary algorithm approach and a new benchmark of images

<https://doi.org/10.1007/s10044-021-01007-6>

Grape Bunch Detection Using A Pixel-Wise Classification In Image Processing - IEEE Conference Publication

<https://doi.org/10.1109/ROPEC50909.2020.9258707>

Method to extract an enhanced cervical vertebrae area from a digital X-ray image

<https://doi.org/10.1016/j.mex.2018.06.009>

Professional References

Dr. Carlos Alberto Brizuela Rodríguez

Researcher, Department of Computer Science, Ensenada Center for Scientific Research and Higher Education (CICESE), Ensenada, B.C.
Email: cbrizuel@cicese.edu.mx

Dr. Miguel Enrique Martínez Rosas

Professor Emeritus, Faculty of Engineering, Architecture and Design, UABC, Ensenada Campus.
Email: emartine@uabc.edu.mx

M.D. Elizabeth Santillán Tarazón

Head of the IT Department, Instituto Tecnológico de Agua Prieta (TecNM/ITAP), Agua Prieta, Sonora.
Phone: +52 (633) 113 6570

Master Ana Silvia Armenta Martínez

Head of Digital Graphic Design Engineering Department.
Email: anasilvia.armenta@cetys.mx
CETYS University, Ensenada Campus
Phone +52 (646) 174-5095 Ext. 3233

Eng. Iván Jesús Martínez

Team Leader of Media Tools Solutions.
Email: ivan.m@samsung.com
Samsung Research Tijuana
Phone +52 (664) 127 1708